

MECH 191 — Metallurgy

Semester Metallography Portfolio

EXAMET-5-R-600-HD · 50–500× Reflected Light Microscopy
Fall 2026 · Leeward Community College · Instructor: Kai Santos

How to Use This Portfolio

This portfolio contains 15 modules — one per lecture week of MECH 191. Each module is completed during or immediately after class using the shared class specimen kit.

For each module you will: (1) check out the kit, (2) examine assigned specimens on the EXAMET-5 at the listed magnifications, (3) complete the observation table, (4) produce a sketch or photograph of the key microstructural feature, and (5) answer the short-answer questions — then return the kit before leaving class.

Completed modules are kept together and submitted as a single portfolio at the end of Week 16.

Rule	Detail
Kit handling	Return ALL specimens before leaving class. Never clean with abrasives. Handle by edges only — fingerprints etch and obscure microstructure.
Damage	If a specimen is damaged, notify the instructor immediately.
PPE	Safety glasses and gloves are required during all lab activities.
Missed sessions	Live demonstrations are not repeated for absences (excused or unexcused).

Student Name: _____ Student ID: _____

Semester / Year: _____ Section: _____

MODULE 1 · WEEK 1 | Atomic Structure & Bonding

Grain boundaries are the 2-D interfaces between adjacent crystals. Their size, shape, and misorientation angle directly control strength, toughness, and ductility — the Hall-Petch relationship states that smaller grains produce stronger metals.

SPECIMENS	A – 1018 Low-carbon steel (normalized, etched)
MAGNIFICATIONS	50× – overview scan 100× – grain boundary network 100× – grain orientation contrast 500× – inclusions (MnS stringers)
ETCHANT	Nital 2% (2% HNO ₃ in ethanol)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD metallurgical microscope
- Pre-mounted Specimen A (1018 steel, phenolic mount)
- Lint-free lens tissue
- Cotton gloves or finger cots
- Ruler/scale for field-width grain estimates
- Pencil and this portfolio handout

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – overview scan → 100× – grain boundary network → 100× – grain orientation contrast → 500× – inclusions (MnS stringers)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundary network visible	50×	☐
Equiaxed grain shape confirmed	100×	☐
Grain orientation contrast (light/dark grains)	100×	☐
MnS inclusion stringers (gray, elongated)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Sketch or describe the grain boundary network at 100×. Are the grains roughly equiaxed or elongated?
Answer	
Q2	Atomic bonding in steel is metallic. What property of metallic bonding allows grain boundaries to exist without the material cracking?
Answer	
Q3	Estimate the average grain diameter at 100× (field width ≈ 2.3 mm on EXAMET-5). Count grains across the field and divide the width by that count.
Answer	
Q4	If you increased to 500×, what new detail appears that was invisible at 100×?
Answer	

MODULE 2 · WEEK 2 | Mechanical Properties

Mechanical properties emerge from microstructure. Pearlite is harder and stronger than ferrite; a finer grain size raises yield strength (Hall-Petch). This module connects what you measure in a test lab to what you see at the microscopic scale.

SPECIMENS	A – 1018 steel (normalized) B – 1045 steel (normalized), Vickers hardness indent applied for Q2
MAGNIFICATIONS	100× – indent identification and measurement 100× – ferrite/pearlite ratio comparison 500× – lamellar pearlite spacing
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B (phenolic mounts, Vickers indent applied to B for Q2)
- Stage micrometer or calibrated eyepiece graticule
- Calculator (for HV formula)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion). 2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – indent identification and measurement → 100× – ferrite/ pearlite ratio comparison → 500× – lamellar pearlite spacing

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
White ferrite grains visible (A vs B comparison)	100×	☐
Dark pearlite colonies visible	100×	☐
Square Vickers indent visible on Specimen B	100×	☐
Both diagonals of indent measurable	100×	☐
Lamellar pearlite lamellae resolvable	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Compare Specimens A and B at 100×. Which has more pearlite? Explain why, given 1045 has more carbon than 1018.
Answer	
Q2	Measure the Vickers indent diagonal on Specimen B (after the indent is applied) at 100×. The field width at 100× is ~2.3 mm. Estimate the diagonal in mm, then calculate: $HV = 1.854 \times F / d_{avg}^2$. (Use $F = 1 \text{ kgf} = 9.81 \text{ N}$; d in mm.)
Answer	
Q3	Hall-Petch: $\sigma_y = \sigma_o + k/\sqrt{d}$. If grain size is refined by 50%, what happens to yield strength? (Qualitative answer is fine.)
Answer	
Q4	Which specimen would you expect to have higher tensile strength — and why? Does the microstructure support that prediction?
Answer	

MODULE 3 · WEEK 4 | Carbon & Alloy Steels

The iron-carbon phase diagram predicts the equilibrium ferrite and pearlite fractions at a given carbon content. Comparing three normalized steels spanning the practical carbon range — low-carbon 1018, medium-carbon 1045, and high-carbon 1075 — shows how composition alone controls the ferrite/pearlite balance, and therefore hardness and strength, without any heat treatment involved.

SPECIMENS	A – 1018 steel (normalized) B – 1045 steel (normalized) C – 1075 steel (normalized)
MAGNIFICATIONS	100× – phase field overview 100× – ferrite vs. pearlite distinction 500× – pearlite lamellar spacing across all three
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B, C (phenolic mounts)
- Iron-carbon phase diagram reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – phase field overview → 100× – ferrite vs. pearlite distinction → 500× – pearlite lamellar spacing across all three

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Predominantly ferrite + minor pearlite (1018)	100×	☐
Roughly equal ferrite/pearlite (1045)	100×	☐
Dominant pearlite with thin ferrite network (1075)	100×	☐
Pearlite lamellae resolvable across all three specimens	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Draw a simple schematic of the iron-carbon phase diagram and mark where 1018, 1045, and 1075 each fall at room temperature after normalizing.
Answer	
Q2	At 100×, rank Specimens A, B, and C by visually estimated pearlite fraction. Does the ranking match what you'd predict from each grade's carbon content using the lever rule?
Answer	
Q3	At 500×, compare pearlite lamellar spacing across the three specimens. Does spacing change meaningfully with carbon content, or is it controlled primarily by cooling rate instead?
Answer	
Q4	1075 has roughly four times the carbon content of 1018. Would you expect 1075 to have four times the hardness? Explain using what you know about how pearlite fraction relates to hardness.
Answer	

MODULE 4 · WEEK 5 | Stainless Steel Families

Stainless steel resists corrosion through a passive chromium oxide layer, but different stainless families reach very different microstructures from that same starting point. 304 and 316 are both austenitic (FCC, non-magnetic) — 316 differs mainly by a molybdenum addition for chloride resistance. 17-4PH is a precipitation-hardening martensitic grade with a completely different strengthening mechanism and microstructure. Comparing all three in one sitting shows that 'stainless' describes a corrosion behavior, not a single microstructure.

SPECIMENS	D – 304 austenitic stainless steel E – 316 austenitic stainless steel F – 17-4PH precipitation-hardening stainless steel
MAGNIFICATIONS	100× – austenite grain structure and twin bands (D, E) 100× – martensitic matrix and precipitate distribution (F) 500× – twin boundary detail / fine precipitate detail
ETCHANT	Marble's reagent (D, E): 10g CuSO ₄ + 50 mL HCl + 50 mL H ₂ O Kalling's No. 2 (F): 5g CuCl ₂ + 100 mL HCl + 100 mL ethanol

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens D, E, F (phenolic mounts)
- PREN / stainless family reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion). 2.

Handle specimen by its edges — fingerprints etch and will obscure the microstructure.

3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.

4. Switch to 100× without moving the stage. Fine-focus only above 100×.

5. Work through magnifications in sequence: 100× – austenite grain structure and twin bands (D, E) → 100× – martensitic matrix and precipitate distribution (F) → 500× – twin boundary detail / fine precipitate detail **Field**

widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed austenite grains, 304 (Specimen D)	100×	☐
Equiaxed austenite grains, 316 (Specimen E)	100×	☐
Straight annealing twin bands crossing multiple grains (D, E)	100×	☐
Martensitic matrix, no twinning (17-4PH, Specimen F)	100×	☐
Fine precipitate distribution (Specimen F)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: _____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Identify annealing twins in Specimens D and E. Why do FCC austenitic stainless steels twin readily during annealing while the martensitic Specimen F does not?
Answer	
Q2	316 adds roughly 2–3% molybdenum compared to 304. That addition barely changes what you can see under the microscope, yet 316 is specified over 304 in marine and chloride-heavy environments. What does Mo actually do, and why doesn't its benefit show up as a visible microstructural difference?
Answer	
Q3	Specimen F (17-4PH) looks nothing like D or E under the scope. Describe what you observe instead, and explain why 'stainless' doesn't guarantee a particular microstructure the way '1018' or '1045' does for carbon steel.
Answer	
Q4	17-4PH is strengthened by precipitation during aging, not by the FCC structure that produces twinning in D and E. Based on what you observed, would you expect 17-4PH to be more or less prone to work hardening during machining than 304? Explain your reasoning.
Answer	

MODULE 5 · WEEK 6 | Nonferrous Metals & Phase Diagrams

Nonferrous alloys rely on different hardening mechanisms than steel. Aluminum uses precipitation hardening; brass relies on solid-solution strengthening. Both show grain structures at similar magnifications to steel but with very different twin densities — a direct consequence of stacking fault energy.

SPECIMENS	G – 6061-T6 Aluminum alloy K – 70/30 Brass (Cu-Zn)
MAGNIFICATIONS	50× – grain structure overview 100× – grain boundaries and twins 500× – twin band detail 500× – Mg ₂ Si precipitates in aluminum
ETCHANT	Keller's reagent (G): 2 mL HF + 3 mL HCl + 5 mL HNO ₃ + 190 mL H ₂ O Ferric chloride (K): 5g FeCl ₃ + 50 mL HCl + 100 mL H ₂ O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens G, K (phenolic mounts)
- Al-Mg-Si precipitation / Cu-Zn stacking-fault reference
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion). 2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – grain boundaries and twins → 500× – twin band detail → 500× – Mg₂Si precipitates in aluminum

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed aluminum grains (6061-T6, Specimen G)	100×	☐
Fine dark Mg ₂ Si precipitate dots	500×	☐
Very prominent annealing twins in brass (Specimen K, almost every grain)	40–100×	☐
Multiple parallel twin bands per grain (brass)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____ × Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Compare twin density in Specimen K (brass) vs. Specimen D (304 SS, Module 4). Both are FCC. Which has more prominent twins and why? (Hint: stacking fault energy.)
Answer	
Q2	The 6061-T6 designation means solution heat treated and artificially aged. What does aging do to the microstructure, and why does it increase hardness?
Answer	
Q3	At 500× on the aluminum specimen, describe any fine particles observed. What are they likely to be, and how do they strengthen the alloy?
Answer	
Q4	Using the Al-Mg-Si phase diagram concept, explain why solution treatment must precede aging. What happens if the solution treatment step is skipped?
Answer	

MODULE 6 · WEEK 7 | Mechanical Testing

Mechanical tests produce numbers; microstructure explains why. Hardness testing leaves a permanent indent whose geometry encodes the hardness value. Grain size is a quantitative microstructural measurement (ASTM E112) that directly correlates to strength and toughness through the Hall-Petch relationship.

SPECIMENS	A – 1018 steel (normalized) — grain size measurement B – 1045 steel (normalized) — grain size comparison, Vickers indent applied for Q1
MAGNIFICATIONS	100× – grain intercept counting (ASTM E112) 100× – Vickers indent diagonal measurement 100× – grain structure comparison A vs B
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD with calibrated eyepiece graticule
- Specimens A, B (phenolic mounts)
- Stage micrometer (to calibrate graticule)
- Calculator, pencil, ruler for test-line intercept counting

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – grain intercept counting (ASTM E112) → 100× – Vickers indent diagonal measurement → 100× – grain structure comparison A vs B

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundaries countable at 100× (intercept method)	100×	☐
Square Vickers indent with both diagonals measurable	100×	☐
Plastic zone around indent (disturbed grain boundaries)	100×	☐
Grain size noticeably different between A and B	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: _____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Measure both diagonals d_1 and d_2 of the Vickers indent placed on Specimen B. Calculate $HV = 1.854 \times F / d_{avg}^2$ ($F = 1 \text{ kgf} = 9.81 \text{ N}$; d in mm). Report your HV result.
Answer	
Q2	Using the ASTM E112 lineal intercept method on Specimen A at $100\times$ (field width $\approx 2.3 \text{ mm}$): draw a test line, count grain boundary intersections, and estimate ASTM Grain Size Number G.
Answer	
Q3	A tensile test on 1045 normalized steel gives $UTS \approx 750 \text{ MPa}$. Using the empirical relation $UTS \approx 3.3 \times HV$, does your measured HV predict a consistent UTS?
Answer	
Q4	Impact toughness (Charpy) is sensitive to grain size. If the grain size of Specimen B were refined by heat treatment, would toughness increase or decrease — and why?
Answer	

MODULE 7 · WEEK 8 | NDT Methods

NDT methods detect discontinuities indirectly — through magnetic flux leakage, ultrasonic reflection, or surface-penetrant bleed-out. Metallography is the destructive confirmation method NDT is designed to avoid — a distinction that matters most for parts too expensive or too critical to sacrifice. This module bridges the two approaches.

SPECIMENS	A – 1018 steel (baseline reference) L – 360 leaded brass (discrete second-phase particles) P – Ti-6Al-4V titanium (aerospace-grade, NDT-first material)
MAGNIFICATIONS	50× – full-section scan for gross defects 100× – discrete particle identification (L) 500× – particle/inclusion morphology detail
ETCHANT	Nital 2% (A) Ferric chloride (L) Kroll's reagent (P)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, L, P (phenolic mounts)
- NDT method comparison reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).

Handle specimen by its edges — fingerprints etch and will obscure the microstructure.

3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full-section scan for gross defects → 100× – discrete particle identification (L) → 500× – particle/inclusion morphology detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Surface or near-surface void/pore located (Specimen A)	40–100×	☐
Elongated MnS inclusion aligned with rolling direction (Specimen A)	100×	☐
Discrete lead particles identified in Specimen L, distinguished from a true defect	100×	☐
General surface condition of Specimen P noted (standard polish, no destructive prep)	50×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	In Specimen L, the lead particles are an intentional alloying addition, not a defect — yet under the scope they look like discrete dark inclusions, similar to how a real defect might appear. Which NDT method — PT, MT, or UT — would best distinguish an engineered particle from an unwanted inclusion in a real part, and why can't NDT alone always make that distinction?
Answer	
Q2	An elongated dark feature aligned with the rolling direction is visible at 100× in Specimen A. Would MT produce a strong or weak indication on this feature in a ferromagnetic part? Explain.
Answer	
Q3	Titanium aerospace components like Specimen P's alloy are almost never sectioned for metallography in service — they're inspected by NDT throughout their service life instead. Given what titanium costs and where it's used, explain why destructive metallography is reserved for failure investigations and process qualification, not routine inspection.
Answer	
Q4	Metallography destroys the part. Give two situations where destructive metallographic examination is justified over NDT in production or maintenance.
Answer	

MODULE 8 · WEEK 9 | Quality Standards & Defect Recognition

Quality standards require documented, reproducible measurements. ASTM E112 for grain size and ASTM E45 for inclusion rating both require systematic microscopic examination with written records — this module practices that discipline.

SPECIMENS	A – 1018 steel — formal ASTM E112 grain size report
MAGNIFICATIONS	100× – all grain-size measurements 100× – inclusion survey (E45)
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD with calibrated graticule
- Specimen A (phenolic mount)
- ASTM E45 inclusion chart comparators (textbook or standard)
- Examination record form (write in this portfolio)
- Calculator, pencil

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).

Handle specimen by its edges — fingerprints etch and will obscure the microstructure.

3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – all grain-size measurements → 100× – inclusion survey (E45)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundaries consistently resolvable (three test lines)	100×	☐
Type A inclusions (sulfide stringers) identified	100×	☐
Type B, C, or D inclusions noted (if present)	100×	☐
Inclusion severity rated Thin or Heavy vs. chart	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: _____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Complete the grain size measurement using ASTM E112 intercept method (three test lines, averaged). Record your result as ASTM Grain Size Number G.
Answer	
Q2	Perform a simplified ASTM E45 inclusion survey at 100×. For each type (A, B, C, D), record Thin or Heavy severity. Present findings in a table.
Answer	
Q3	A material certificate states 'ASTM Grain Size 7 minimum.' Does your specimen meet this requirement? (G7 ≈ 32 μm grain diameter.)
Answer	
Q4	A quality inspector must document a metallographic examination per written procedure. List four data items that must appear in a conforming examination record.
Answer	

MODULE 9 · WEEK 10 | Wrought Aluminum Alloy Systems

Not every aluminum alloy is strengthened the same way. 6061 is an Al-Mg-Si (6xxx) alloy, precipitation-hardened by fine Mg_2Si particles; 2024 is an Al-Cu (2xxx) alloy, precipitation-hardened by a different phase with coarser, more visible constituent particles. Comparing both wrought alloys side by side shows how two grades can share a strengthening mechanism — aging — while producing very different microstructures and very different corrosion behavior.

SPECIMENS	G – 6061 aluminum alloy J – 2024 aluminum alloy
MAGNIFICATIONS	50× – grain structure overview 100× – constituent particle survey 500× – fine precipitate and constituent particle detail
ETCHANT	Keller's reagent: 2 mL HF + 3 mL HCl + 5 mL HNO_3 + 190 mL H_2O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens G, J (phenolic mounts)
- Stage micrometer or calibrated graticule (for constituent particle size estimate)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – constituent particle survey → 500× – fine precipitate and constituent particle detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed grain structure visible in both specimens	50×	☐
Constituent particles identified in Specimen G (6061)	500×	☐
Constituent particles identified in Specimen J (2024)	500×	☐
Relative particle size/density compared between G and J	500×	☐
Grain boundary and twin structure noted in both	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 500×, compare the size and number of visible constituent particles between Specimens G and J. Which alloy shows coarser, more numerous particles, and does that match what you'd expect from Al-Cu vs. Al-Mg-Si intermetallic chemistry?
Answer	
Q2	Both alloys are strengthened by aging, yet 2024 is notably less corrosion-resistant than 6061 in service (it's often sold clad with pure aluminum for exactly this reason). Based on the alloying elements each contains, propose a reason why.
Answer	
Q3	If you swapped the aging treatment of these two alloys — 2024's schedule onto 6061's composition and vice versa — would you expect similar strengthening in both? Why or why not?
Answer	
Q4	6061 is a general-purpose structural alloy; 2024 is an aerospace skin alloy. Based on the microstructures you observed, suggest one property difference — beyond corrosion resistance — that would drive an engineer to choose one over the other.
Answer	

MODULE 10 · WEEK 11 | Aluminum Alloys for Forming Applications

Alloy selection for a forming process is driven as much by how an alloy responds to deformation as by its final strength. 6063 is formulated specifically for extrudability and surface finish — the standard choice for architectural window and door sections. 5052 is a non-heat-treatable Al-Mg alloy prized for its formability in sheet stampings and its corrosion resistance in marine environments. Comparing the two shows how composition is chosen around a manufacturing process, not just a target strength.

SPECIMENS	H – 6063 aluminum alloy I – 5052 aluminum alloy
MAGNIFICATIONS	50× – grain structure overview 100× – grain boundary and second-phase comparison 500× – fine microstructural detail
ETCHANT	Keller's reagent

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens H, I (phenolic mounts)
- Ruler for field-width measurement
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – grain boundary and second-phase comparison → 500× – fine microstructural detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed grain structure visible in both specimens	100×	☐
Fine Mg ₂ Si precipitates identified in Specimen H (6063)	500×	☐
Absence of precipitates confirmed in Specimen I (5052, solid-solution only)	500×	☐
Relative grain size compared between H and I	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: _____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	6063 is heat-treatable (Mg_2Si precipitates); 5052 is not (Mg stays in solid solution). At 500×, describe what you can and cannot see in each specimen that reflects this difference.
Answer	
Q2	6063 is formulated specifically for extrusion — lower alloy content than 6061 for smoother die flow and a better surface finish. Based on your observations, what about its microstructure would make it easier to extrude than a more heavily alloyed grade?
Answer	
Q3	5052 work-hardens rather than age-hardens. If a sheet of 5052 were cold-stamped and then examined at 100×, what microstructural change would you expect to see compared to the as-received specimen?
Answer	
Q4	A designer needs a formed aluminum bracket for a marine application. Based on the strengthening mechanism of each alloy, which would you recommend — 6063 or 5052 — and why?
Answer	

MODULE 11 · WEEK 12 | Machining Processes

Machining removes material by chip formation, but the cutting zone generates heat and stress that can alter near-surface microstructure. Surface finish and subsurface integrity are quality-critical: a ground surface that looks perfect to the eye may carry thermal damage detectable only under the microscope.

SPECIMENS	A – 1018 steel (baseline edge condition) B – 1045 steel (machined edge) G – 6061 aluminum (machined edge comparison)
MAGNIFICATIONS	50× – edge overview 500× – near-surface grain deformation 500× – surface features / preparation scratches
ETCHANT	Nital 2% (A, B) Keller's reagent (G)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B, G
- Surface roughness comparator card (optional reference)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – edge overview → 500× – near-surface grain deformation → 500× – surface features / preparation scratches

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Prepared edge visible and smooth at 50×	50×	☐
Near-surface grain distortion or deformation layer (steel)	500×	☐
Preparation scratches visible at 500×	500×	☐
Comparison of near-surface condition: steel vs. aluminum (Specimen G)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 500×, examine the prepared edge of Specimen A. Does the near-surface grain structure differ from the interior? Describe any deformation layer you observe.
Answer	
Q2	In machining, tool wear increases cutting forces and heat. What microstructural change could this cause in a high-carbon steel workpiece — and how would it appear at 500×?
Answer	
Q3	Surface finish affects fatigue life. Identify one feature visible at 500× on any specimen that would act as a fatigue crack initiation site on a machined surface in service.
Answer	
Q4	Compare the near-surface condition of steel and aluminum specimens. Which shows more preparation-induced deformation? Why might aluminum be more susceptible?
Answer	

MODULE 12 · WEEK 13 | Copper Alloys: Strengthening Mechanisms

Powder metallurgy and additive manufacturing are covered in this week's lecture material; this module's hands-on specimen instead continues the strengthening-mechanism thread using pure and alloyed copper. Specimen N is commercially pure copper — no alloying, no precipitates, just grain boundaries and annealing twins. Specimen O is chromium copper, a precipitation-hardening alloy that trades a little electrical conductivity for a real strength increase — the same aging mechanism you've now seen in steel (17-4PH) and aluminum (2024), applied to a third base metal entirely.

SPECIMENS	N – 110 copper (commercially pure) O – 182 chromium copper (precipitation-hardening)
MAGNIFICATIONS	50× – grain structure overview 100× – twin band comparison 500× – precipitate detail (O)
ETCHANT	Ferric chloride: 5 g FeCl ₃ + 50 mL HCl + 100 mL H ₂ O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens N, O (phenolic mounts)
- Eyepiece graticule with grid (for twin/precipitate density comparison)
- Cross-family precipitation hardening summary card (17-4PH, 2024, chromium copper)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – twin band comparison → 500× – precipitate detail (O) **Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm**

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed grain structure visible, Specimen N (pure copper)	100×	☐
Annealing twin bands visible in Specimen N	100×	☐
Fine precipitate distribution identified in Specimen O (chromium copper)	500×	☐
Grain/twin structure compared between N and O	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____ × Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 100×, compare grain and twin structure between Specimens N and O. Which looks 'simpler,' and why would you expect that from a single-element metal versus an alloyed one?
Answer	
Q2	You've now seen precipitation hardening in three different base metals this semester: 17-4PH (steel), 2024 (aluminum), and Specimen O (copper). List the strengthening phase in each, and state what all three approaches have in common at the mechanism level.
Answer	
Q3	Pure copper (Specimen N) has excellent electrical conductivity; the chromium addition in Specimen O reduces that conductivity somewhat. Why does adding any alloying element — even in small amounts — generally reduce electrical conductivity in a metal?
Answer	
Q4	Powder metallurgy (this week's lecture topic) sinters metal powder into a part without ever fully melting it. Based on the wrought, fully-dense specimens you've examined all semester, what microstructural feature would you expect to see in a PM part that you have not seen in any specimen this semester?
Answer	

MODULE 13 · WEEK 14 | Brass Alloys: Composition and Microstructure

Welding process fundamentals are covered in this week's lecture material; this module's hands-on specimen instead examines how zinc content changes brass microstructure across three grades. Specimen K (cartridge brass, 70/30 Cu-Zn) is single-phase alpha. Specimen L (free-machining brass) adds a small lead addition for machinability. Specimen M (naval brass) crosses into a two-phase alpha-beta structure at higher zinc content. Together they show a single compositional variable — zinc — driving a real change in phase count, not just grain size.

SPECIMENS	K – 260 brass (cartridge, single-phase alpha) L – 360 brass (leaded, free-machining) M – 464 brass (naval, duplex alpha-beta)
MAGNIFICATIONS	100× – single-phase grain structure and twins (K) 100× – lead particle distribution (L) 100× – alpha/beta phase identification (M) 500× – phase boundary detail (M)
ETCHANT	Ferric chloride: 5 g FeCl ₃ + 50 mL HCl + 100 mL H ₂ O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens K, L, M (phenolic mounts)
- Cu-Zn (brass) phase diagram reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – single-phase grain structure and twins (K) → 100× – lead particle distribution (L) → 100× – alpha/beta phase identification (M) → 500× – phase boundary detail (M)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed single-phase alpha grains, Specimen K (260)	100×	☐
Annealing twin bands visible in Specimen K	100×	☐
Discrete lead particles identified, Specimen L (360)	100×	☐
Two distinct phases (alpha + beta) identified, Specimen M (464)	100×	☐
Alpha/beta phase boundary morphology noted, Specimen M	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: _____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 100×, identify annealing twins in Specimen K. Are they still visible in Specimens L and M, or does the added lead/beta phase make them harder to see?
Answer	
Q2	Locate the lead particles in Specimen L. Lead is essentially insoluble in the copper-zinc matrix. Explain how that insolubility is exactly what makes it useful for machinability.

Answer	
Q3	Specimen M is the only one of the three showing two distinct phases (alpha and beta) rather than one. Based on the zinc contents of these three brasses, what does that tell you about where the alpha/beta phase boundary falls on the Cu-Zn phase diagram?
Answer	
Q4	A component needs to be both machined economically and resistant to dezincification in seawater. Based on what you observed in Specimens K, L, and M, would you specify 360 or a duplex naval brass — and what's the microstructural tradeoff?
Answer	

MODULE 14 · WEEK 15 | Precipitation Hardening: A Cross-Family Comparison

Weld metallurgy and failure analysis are covered in this week's lecture material. This module instead revisits three specimens examined earlier in the semester — Specimen F (17-4PH stainless steel, Module 4), Specimen J (2024 aluminum, Module 9), and Specimen O (182 chromium copper, Module 12) — to directly compare how the same strengthening mechanism, precipitation hardening, plays out across three different base metals, crystal structures, and precipitate chemistries.

SPECIMENS	F – 17-4PH stainless steel (revisit, Module 4) J – 2024 aluminum (revisit, Module 9) O – 182 chromium copper (revisit, Module 12)
MAGNIFICATIONS	500× – precipitate / constituent particle comparison across all three
ETCHANT	As previously used — Kalling's No. 2 (F), Keller's reagent (J), ferric chloride (O). No re-etch required if specimens are still in prepared condition.

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens F, J, O (revisit — retrieve from kit, phenolic mounts)
- Calibrated graticule (precipitate size comparison)
- Precipitation hardening cross-family summary card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 500× – precipitate / constituent particle comparison across all three (F, J, O)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Precipitate morphology re-examined in Specimen F (17-4PH)	500×	☐
Precipitate morphology re-examined in Specimen J (2024)	500×	☐
Precipitate morphology re-examined in Specimen O (chromium copper)	500×	☐
Relative precipitate size/density ranked across all three	500×	☐
Base crystal structure noted for each (martensitic vs. FCC)	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____ × Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Fill in a short table from memory and your earlier observation notes: for each of F, J, and O, name the base metal, the precipitate phase (if known), and whether it appeared fine/hard-to-resolve or coarse/clearly visible at 500×
Answer	
Q2	All three of these alloys go through the same basic aging heat-treatment sequence conceptually: solutionize, quench, age. Explain in your own words what each of those three steps accomplishes, using one of the three specimens as your example.
Answer	
Q3	Precipitation hardening increases strength without changing the base crystal structure (martensitic in F, FCC in J and O). Why is a fine, closely-spaced precipitate more effective at blocking dislocation motion than a coarse, widely-spaced one?
Answer	
Q4	If all three specimens were left at room temperature for years ('natural aging') instead of artificially aged, would you expect them to reach the same final strength as the artificially aged condition? Explain why or why not, using what you know about diffusion rates at room temperature.
Answer	

MODULE 15 · WEEK 16 | Course Capstone & Synthesis

Corrosion mechanisms and protection methods are covered in this week's lecture material and the course reference pages — including a direct callback to Specimens D and E (304 vs. 316) from Module 4, since molybdenum content and PREN are exactly why 316 outperforms 304 in chloride environments. This final module is a synthesis exercise: no new specimen, just your own observations from across the semester, organized into one final comparison.

SPECIMENS	No new specimen — synthesis exercise using observation notes and sketches from Modules 1–14
MAGNIFICATIONS	N/A — reference your own notes and sketches from prior modules
ETCHANT	N/A

EQUIPMENT & SUPPLIES

- Portfolio notes, sketches, and answered questions from Modules 1–14
- PREN / corrosion reference card (for the Module 4 callback)
- Precipitation hardening summary from Module 14
- Pencil, calculator
- This portfolio, for final review

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Gather your observation tables, sketches/photographs, and answered questions from Modules 1–14.
2. Re-read your notes on Specimens D and E (304 vs. 316, Module 4) before answering the PREN-related question.
3. Re-read your notes on Specimens F, J, and O (Module 4, 9, 12) before answering the precipitation-hardening synthesis question.
4. No microscope session is required for this module — it is a written synthesis exercise.
5. Review the full course arc (Units 1–5) before writing your synthesis answer.

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Crystal structure identified for at least one BCC, FCC, and HCP specimen (from notes)	—	☐
Twin presence/absence compared across at least three specimens (from notes)	—	☐
Strengthening mechanism identified for at least three specimens (from notes)	—	☐
Precipitation-hardening thread traced across F, J, O (from notes)	—	☐
Five-unit synthesis chain outlined with specimen citations	—	☐

STEP 2 — Sketch or Photograph

Specimen: _____ Magnification: ____× Etchant: _____

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Across every specimen you examined this semester, identify one example each of a BCC structure, an FCC structure, and an HCP structure (Specimen P, titanium, Module 7). List the specimen and family for each.
Answer	
Q2	You observed annealing twins in several FCC specimens (304, 316, brass) but never in the BCC carbon steels. Explain why, in terms of stacking fault energy and crystal structure, twinning is common in some metals and essentially absent in others.
Answer	
Q3	You also encountered three strengthening mechanisms this semester: solid-solution strengthening (5052, pure copper — no precipitates), precipitation/age hardening (17-4PH, 2024, chromium copper), and grain-boundary strengthening via Hall-Petch (present to some degree in every specimen). For any one specimen, identify which mechanism(s) applied and explain your reasoning.
Answer	
Q4	Connect the course's five units using your own microscopy observations as evidence: How does atomic structure/bonding (Unit 1) — alloy family and composition (Unit 2) — mechanical testing and inspection (Unit 3) — manufacturing process and alloy selection (Unit 4) — service behavior including corrosion resistance (Unit 5) form a single chain? Cite at least three specimens by letter as evidence.
Answer	

Appendix A — Class Specimen Kit & Field Reference

The kit contains 16 pre-prepared, labeled metallographic specimens. Do not regrind or re-etch. Return all specimens in the labeled foam tray.

Magnification Field Widths — EXAMET-5-R-600-HD

Magnification	Field Width (EXAMET-5)	Ideal for
50×	≈ 8.0 mm	Full cross-section overview, gross specimen scan
100×	≈ 2.3 mm	Grain counting (ASTM E112), inclusion survey (ASTM E45), indent identification, grain aspect ratio, ferrite/pearlite and phase fraction comparison
500×	≈ 0.46 mm	Precipitates and constituent particles, twin boundary detail, phase boundary detail (duplex brass, precipitation-hardening grades), pearlite lamellar spacing

Specimen Summary

Specimen	Material	Mount	Etchant
A	1018 steel (normalized)	Hot-compression phenolic	Nital 2%
B	1045 steel (normalized)	Hot-compression phenolic	Nital 2%
C	1075 steel (normalized)	Hot-compression phenolic	Nital 2%
D	304 austenitic stainless steel	Hot-compression phenolic	Marble's reagent
E	316 austenitic stainless steel	Hot-compression phenolic	Marble's reagent
F	17-4PH precipitation-hardening stainless	Hot-compression phenolic	Kalling's No. 2
G	6061 aluminum alloy	Hot-compression phenolic	Keller's reagent
H	6063 aluminum alloy	Hot-compression phenolic	Keller's reagent
I	5052 aluminum alloy	Hot-compression phenolic	Keller's reagent
J	2024 aluminum alloy	Hot-compression phenolic	Keller's reagent
K	260 brass (cartridge, 70/30)	Hot-compression phenolic	Ferric chloride
L	360 brass (leaded, free-machining)	Hot-compression phenolic	Ferric chloride
M	464 brass (naval, duplex alpha-beta)	Hot-compression phenolic	Ferric chloride
N	110 copper (commercially pure)	Hot-compression phenolic	Ferric chloride
O	182 chromium copper (precipitation-hardening)	Hot-compression phenolic	Ferric chloride
P	Titanium Grade 5 (Ti-6Al-4V)	Hot-compression phenolic	Kroll's reagent

Grading Summary

Each module is graded on four components: Observation table completeness, sketch/photograph quality, short-answer accuracy, and kit return/handling. The 15 weekly modules together are worth 75 points; the holistic end-of-semester portfolio review is worth 25 points — totaling 100 points for the semester.

Component	Points per Module	Notes
Observation table	1	Credit for each feature located and checked
Sketch / photograph	1	Labeled with specimen, magnification, etchant
Short-answer questions (4)	2	0.5 pts each; methodology weighted over final number
Kit return & handling	1	On time, no damage, handled by edges
Total per module	5	
15 modules total	75	5 pts × 15 modules
End-of-semester portfolio review	25	Holistic; all 15 modules as a complete document
TOTAL	100	